

मध्यप्रदेश स्टेट इलेक्ट्रॉनिक्स डेव्हलपमेंट कार्पोरेशन लिमिटेड Madhya Pradesh State Electronics Development Corporation Limited

(म.प्र. सरकार का उपक्रम)
(A Govt. of M.P. Undertaking)

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भोपाल, दिनांक: 28/07/2025

प्रति,

सदस्य सचिव,
मध्यप्रदेश राज्य वेटलैंड प्राधिकरण,
पर्यावरण परिसार, ई-5, अरेरा कॉलोनी
भोपाल (म.प्र.) - 462004

विषय: म.प्र. वेटलैंड पोर्टल में GIS आधारित भोज वेटलैंड एवं रामसर साईट की मॉनिटरिंग हेतु GIS आधारित समाधान।

संदर्भ: म. प्र. राज्य वेटलैंड प्राधिकरण का पत्र क्र. 10055/MPSWA/EPCO/2025, दिनांक: 21/03/25

विषयान्तर्गत संदर्भित पत्र का अवलोकन करने का कष्ट करें। इस संबंध में श्री लोकेन्द्र ठक्कर जी एवं टीम से चर्चानुसार एक Techno Commercial प्रस्ताव तैयार किया गया है।

प्राक्कलन के अनुसार वर्णित कार्य के लिए रिसोर्स की अनुमानित कुल राशि रु 32,64,483 + (18% GST) = 38,07,250/- है। ड्रोन सर्वे की लागत वास्तविक आधार पर तय की जाकर कार्य उपरांत प्रेषित की जाएगी।

कृपया उक्त कुल राशि का 50% राशि रु 15,60,974 + (18% GST) = 19,03,625/- का भुगतान निगम (MPSEDC) को करने हेतु सम्बंधित को निर्देशित करने का कष्ट करें।

विस्तृत प्राक्कलन परिशिष्ट (अ) पर संलग्न है एवं भुगतान हेतु निगम का बैंक विवरण परिशिष्ट "ब" पर सुलभ संदर्भ हेतु संलग्न है

उपरोक्त प्रस्ताव में Security Audit/ WhatsApp/e-sign/Drone/SMS/Application का Support एवं Maintenance इत्यादि पर होने वाला व्यय सम्मिलित नहीं है, उक्त व्यय का भुगतान वास्तविक व्यय अनुसार सूचित किया जाएगा।

संलग्न : - उपरोक्तानुसार


(आशीष वशिष्ठ)
प्रबंध संचालक



Techno-Commercial Proposal for
Monitoring of Bhoj Wetland
for
MP Wetland Authority, EPCO



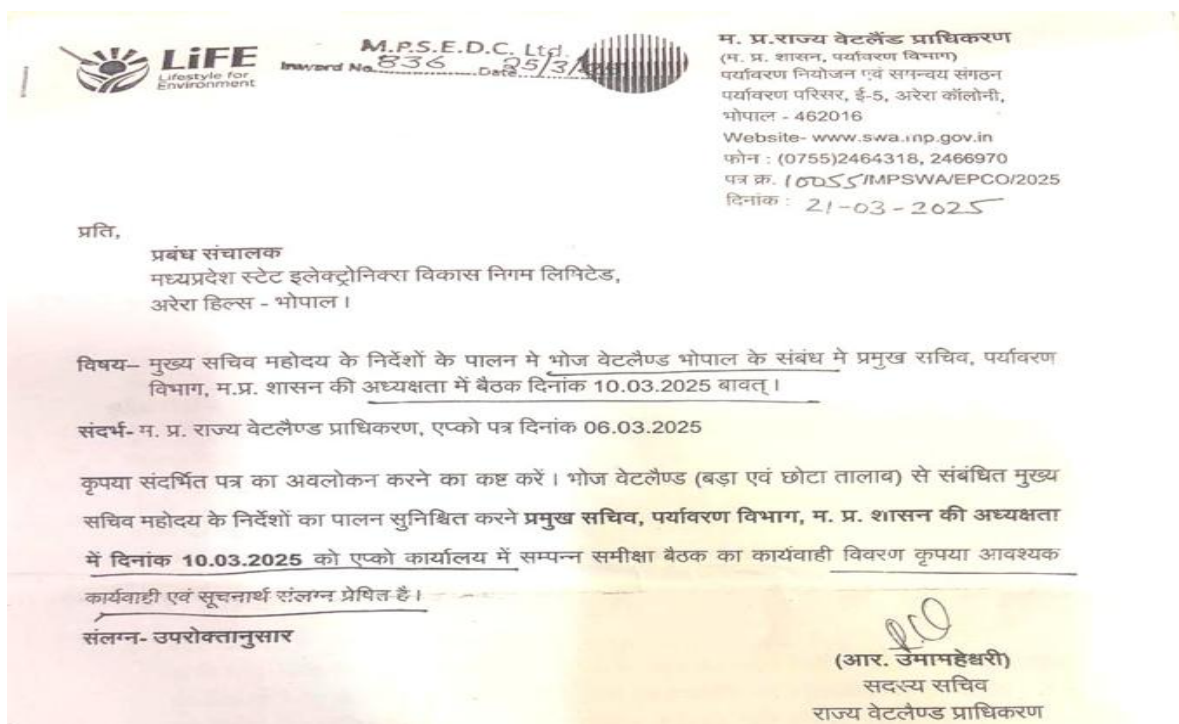
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State IT Center, 47-A, Arera Hills, Bhopal – 462011, Phone: +91-755-2518300, Fax: +91-755-2579824
URL: www.MPSEDC.gov.in, Email: info@MPSEDC.gov.in

Contents

1. INTRODUCTION	3
2. BACKGROUND	3
3. ABOUT EPCO/MPSWA	4
4. ABOUT MPSEDC.....	4
5. OBJECTIVE OF THE PROJECT	4
6. SCOPE OF WORK	5
METHODOLOGY AND APPROACH:	6
1. DEVELOPMENT CYCLE	7
REQUIREMENT COLLECTION AND ANALYSIS:	8
FEASIBILITY STUDY:	8
DESIGN:	8
CODING:	8
TESTING:	8
INSTALLATION/DEPLOYMENT:	9
MAINTENANCE:	9
DELIVERABLES.....	9
PROJECT COSTING	10
DRONE SURVEY COST	11
PAYMENT PLAN	11
EXPECTATIONS & DEPENDENCIES FROM THE DEPARTMENT	12
MPSEDC & EPCO DEPARTMENT DETAILS:	13

1. Introduction

The Bhopal Wetland conservation project is being undertaken by the State Wetland Authority. In a multi-departmental meeting held on 10.03.2025, key stakeholders, including representatives from the Revenue, Forest, Urban Development, and Environment Departments, discussed pressing issues such as encroachment, land use, and environmental impacts. The meeting resulted in strategic decisions aimed at protecting the wetland ecosystem through the strict implementation of the Wetland Rules 2017, declaration of a Zone of Influence (Zoi), updated land classifications, integrated monitoring via GIS tools, and regulated development aligned with environmental sustainability.



2. Background

The Bhopal Wetland conservation project is being undertaken as per the directives of the Principal Secretary, Housing and Urban Development, under the State Wetland Authority. In a multi-departmental meeting held on 10.03.2025, key stakeholders, including representatives from the Revenue, Forest, Urban Development, and Environment Departments, discussed pressing issues such as encroachment, land use, and environmental impacts. The meeting resulted in strategic decisions aimed at protecting the wetland ecosystem through the strict implementation of the Wetland Rules 2017, declaration of a Zone of Influence (Zoi), updated land classifications, integrated monitoring via GIS tools, and regulated development aligned with environmental sustainability.

3. About EPCO/MPSWA

The Environmental Planning & Coordination Organization (EPCO) was established by the Housing and Environment Department of the Government of Madhya Pradesh in 1981. Over the years, EPCO has steadily grown to become the State's premier organization in the field of environment-related matters. It has worked closely with the State Government on various projects, yet it has established its own identity as an autonomous organization. It is a think-tank for environmental matters but is also project-oriented.

The Executive Director is the executive head of the organization. Environmental Planning and Coordination Organization (EPCO) is a unique organization in terms of the foresight and the width of its mandate. In the last 27 years, it has also witnessed rapid developments in the field of environment. Over the last 27 years, EPCO has touched upon virtually every kind of work that is related to the field of the environment. It has carried out pioneering work in the creation of environmental awareness and has focused attention on the environmental problems of the State.

4. About MPSEDC

MPSEDC is the corporation under the Companies Act 1956, which has been entrusted with the responsibility to develop Indore, Bhopal, Jabalpur & Gwalior STP under the guidance of the Department of Science & Technology, Government of Madhya Pradesh. It is tirelessly working towards Promotion, Implementation, and Investment in IT, ITes & ESDM sectors, to propel the growth of Information Technology (IT) in Madhya Pradesh and implement the State IT Policy. MPSeDC is a nodal agency of the state for implementing and providing Remote Sensing, GIS, and Drone-based solutions to government departments.

5. Objective of the project

The objective is to protect the Bhopal Wetland and its surrounding areas by restricting unauthorized activities within the Zone of Influence (ZoI) and Full Tank Level (FTL) by EPCO and other related government departments. It includes strict implementation of the Integrated Management Plan (IMP), updating land classification, and ensuring land use changes are as per wetland and urban development regulations. To identify the change of Land Use / Land Cover, potential environmental threats, provide scientific data, data analysis for the development of sustainable management strategies and policy-making. A web GIS based decision support system with integrated mobile application shall be developed to support field inspections by departmental officials for identifying encroachments, monitoring changes, and improving coordination among departments for effective conservation.

Protection and Preservation: To safeguard the Bhopal Wetland and its associated water sources by preventing encroachments and unauthorized activities within the Zone of Influence (Zoi) and Full Tank Level (FTL) areas.

Regulated Development: To ensure all construction and development activities around the wetland comply with the Wetland Rules 2017, through strict implementation of the Integrated Management Plan (IMP).

Tech-Driven Monitoring: Use RS-GIS tools, drones, satellite imagery, and analytics for continuous oversight.

Updated Land Use Management: To update and align land classification and land use records around the wetland as per environmental and urban development regulations.

Monitoring and Surveillance: To establish a robust monitoring system using GIS tools, satellite imagery, and historical data for continuous assessment and management of the wetland.

6. Scope of work

The scope of work for this project is mentioned below:

- Development of GIS-Enabled Mobile App
- Integration with the wetland portal
- Web portal enables GIS tools to analyze the Wetland using satellite imagery.
- User management
- MIS Report
- User Training
- Conduct drone surveys during **pre-monsoon and monsoon** periods and prepare DSM/DTM as per EPCO request. Drone image resolution will be 5 cm.
- Land Use change identification from satellite imagery (LISS-IV) and Drone imagery, if available.
- A0 (A zero) size map will be prepared as per the department's requirement.
- **Two dedicated resources** — A GIS Data Specialist and a Senior GIS Executive — will be assigned to the project for a duration of one year at the MPSEDC office.

Mobile Application should have the following functionalities:

1. Registration
2. OTP-based login
3. Geo tagging
4. Offline survey module
5. Geofencing
6. Traversing

7. Geotag Photo
8. PDF Upload less than 5 MB

Methodology and Approach:

1. GIS-Enabled Mobile Application for Field Inspections (Ground Truthing)

A customized mobile application integrated with GPS and GIS capabilities to support real-time, on-site data collection by field inspectors/users. Key functionalities include:

- **Geotagged observations:** Record field data with location coordinates.
- **Photographic evidence:** Capture images of encroachments, vegetation, and water levels.
- **Offline data collection:** Enable use in areas with poor connectivity.
- **Instant reporting:** Upload data directly to the central database for further analysis.

2. Drone

a. Drone Imagery Satellite-Based

Conduct drone surveys during **pre-monsoon and monsoon** periods to detect **micro-level changes**, such as:

- Surface water extent fluctuations
- Vegetation cover variations
- Local encroachments or physical disturbances

b. Satellite Imagery (ISRO Data)

Utilize high-resolution **monthly satellite images** (LISS-IV) to monitor **macro-level changes**, including:

- Seasonal shifts in water spread area
- Changes in surrounding land use and cover
- Long-term degradation or recovery patterns

3. Integrated Web GIS Platform – MP Wetland Portal

Integration of all data includes field survey, satellite image, drone image to a **centralized Wetland Portal** to consolidate and visualize all collected data, ensuring seamless access for stakeholders and decision-makers. The portal will:

- User Role Management
- Sync with mobile app and drone data
- Offer layered visualization of **field observations, drone imagery, and satellite data**
- Allow users to **view temporal changes** and generate reports

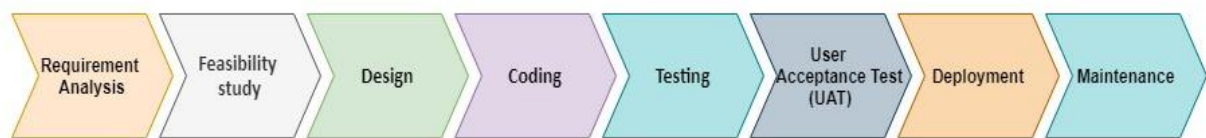
4. Advanced GIS-Based Analytical Tools

Incorporate spatial analysis tools into the portal to enhance decision support through:

- **Change detection** in land use and water extent over time
- **Custom Search**
- **Proximity**
- **Routing**

1. Development Cycle

The development of a Mobile App and web-based GIS portal is divided into the following stages:



Development stages

- Phase 1: Requirement collection and analysis
- Phase 2: Feasibility study:
- Phase 3: Design:
- Phase 4: Coding:
- Phase 5: Testing:
- Phase 6: User Acceptance Test (UAT)
- Phase 7: Installation/Deployment:
- Phase 8: Maintenance:

Requirement collection and analysis:

The requirement is the first stage in the process. In this process, the MPSEDC team will get input from EPCO department officials. Planning for the quality assurance requirements and recognition of the risks involved is also done at this stage.

This stage gives a clearer picture of the scope of the entire project and the anticipated issues, opportunities, and directives that triggered the project.

The requirements-gathering stage needs teams to get detailed and precise requirements.

Feasibility study:

Once the requirement analysis phase is completed the next step is to define and document software needs. This process is conducted with the help of the 'Functional Requirement Specification' document also known as the FRS document. It includes everything that should be designed and developed during the project life cycle.

Design:

In this third phase, the system and software design documents are prepared as per the requirement specification document. This helps define the overall system architecture. This design phase serves as input for the next phase of the model.

Coding:

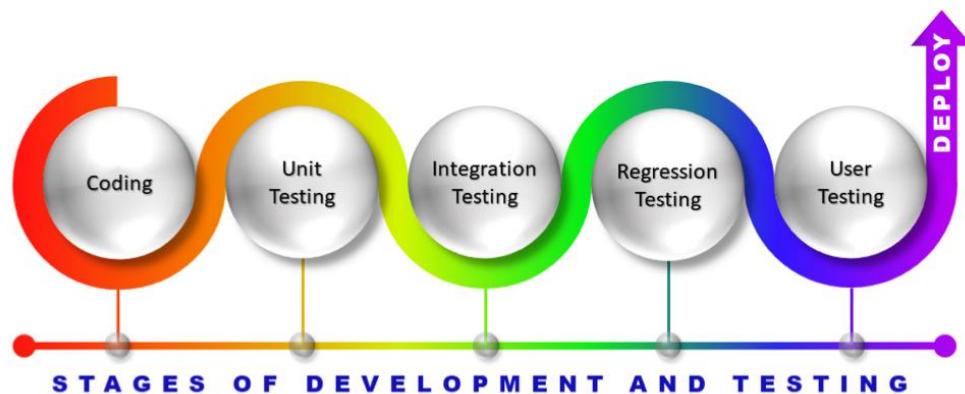
Once the system design phase is over, the next phase is coding. In this phase, developers start building the entire system by writing code using the chosen programming language. In the coding phase, tasks are divided into units or modules and assigned to the various developers. It is the longest phase of the Software Development Life Cycle process.

In this phase, the Developer needs to follow certain predefined coding guidelines. They also need to use programming tools like compilers, interpreters, and debuggers to generate and implement the code.

Testing:

Once the software is complete, it is deployed in the testing environment. The testing team starts testing the functionality of the entire system. This is done to verify that the entire application works according to the client's requirements.

During this phase, the QA and testing team may find some bugs/defects that they communicate to developers. The development team fixes the bug and sends it back to QA for a re-test. This process continues until the software is bug-free, stable, and working according to the business needs of that system.



Software testing stages

Installation/Deployment:

Once the software testing phase is over and no bugs or errors are left in the system, then the final deployment process starts. Based on the feedback given by the project manager, the final software is released and checked for deployment issues if any.

Maintenance:

Once the system is deployed, and clients start using the developed system, the following 3 activities occur

- Bug fixing - bugs are reported because of some scenarios that are not tested at all
- Upgrade - Upgrading the application to newer versions of the Software
- Enhancement - Adding some new features to the existing software

The main focus of this phase is to ensure that needs continue to be met and that the system continues to perform as per the specifications mentioned in the first phase.

Deliverables

Mobile & Web Application – Updated Web application (wetland portal) with new tools and integrated with mobile app as outlined in the above scope.

MIS Report – The report module should be provided as per the provided template by EPCO on the web portal and will be available for download in both Excel and PDF formats.

Drone Data – Integrate drone image and DSM/DTM in the wetland portal.

User Manual – A detailed user manual for both the mobile and web applications.

URM – This module will give access to EPCO for the onboarding of officers.

Map – A0 size map in pdf format.

Two dedicated resources — A GIS Data Specialist and a Senior GIS Executive — will be assigned to the project for a duration of one year for analyzing the data as per EPCO

requirements. Both resources should work at the MPSEDC office under the guidance of the EPCO and MPSEDC officials.

Resource Costing of Project

Resource & Cost Estimation- MP Wetland Authority, EPCO							
S.N.	Resource Type Development & Support Team	Project wise Resource Count	Total Man days (In Months)	CTC (Rs in lac) Y1	Monthly CTC	Total Expenditure	Actual Cost to department
1	Tech Lead	1	1	Shared	Shared	Shared	Shared
2	Business Analyst	1	1	1000000	83333	83333	83333
3	GIS Developer	1	3	1913000	159417	478250	478250
4	Database Administrator	1	2	2200000	183333	366667	366667
5	Database Developer	1	1	1000000	83333	83333	83333
6	Mobile Developer	1	3	1000000	83333	250000	250000
7	Dot Net Developer (angular)	1	2	1000000	83333	166667	166667
8	Software Tester	1	1	1000000	83333	83333	83333
9	UI/UX	1	1	1000000	83333	83333	83333
10	GIS Data Specialist	1	12	780000	65000	780000	780000
11	Sr. GIS Executive	1	12	420000	35000	420000	420000
A	Total (A)			11313000	942750	2794917	2794917
B	HR Expenses						138250
C	Total (B)						2933167
D	MPSEDC Service Charges @5%						146658
E	Contingency Charges @5%						146658
F	Total (C)						3226483
G	GST @ 18 %						580767
H	Grand Total						3807250

Note:

The project cost covers development, testing, UAT, and final deployment.

Two dedicated resources — a GIS Data Specialist and a Senior GIS Executive.

All shared and dedicated team members should work at the MPSEDC office.

1. Costing Assumptions

- The project scope is based on the initial requirements-gathering sessions and is subject to change only upon mutual agreement between the EPCO Department and the MPSEDC team.
- Any future updates or advancements in technology that may impact the solution are assumed to be manageable through regular maintenance and support services.
- Any specific hardware costs (If Needed) shall be based on current market prices and will be charged separately.
- The Cost of Security Audit, SMS, and WhatsApp is not considered in the proposal, and if applicable, shall be as per actuals.

Drone Survey Cost

Description of Work/Item	INR Excluding GST	Unit
Drone Survey	250	per hectare
Data Processing	225	per hectare
Data Extraction	225	per hectare
Total	700	per hectare

*Minimum order size for area-based mapping is 100 hectares.

**KML/.shp file of Ramsar site will be provided by the EPCO.

Payment Plan

S.No.	Payment Type	Payment Description	% of Amount	Total Amount (INR)	Remarks
1	Resource Cost	Advance for Project Initiation	50%	19,03,625	Total Resource Cost: 38,07,250 (INR)
		UAT of Web Portal and Mobile APP	30%	11,42,175	
		Go live of all Modules	20%	7,61,450	
2	Drone Survey	Drone Survey	100% With the request to fly the drone.	It should be calculated based on drone survey area at the Rs. 700 per hectare.	- It is proposed that Drone surveys will be done twice a year—once before the monsoon and once during the monsoon—after getting a request from EPCO. - The department will inform 15 days before the drone survey. The cost will be calculated after the survey, depending on the area covered.
		Data Processing			
		Data Extraction			

*Drone survey cost will be calculated based on actual survey area after conducting the survey and accordingly the amount will be charged.

- SMS charges will be applicable as per the actual.
- Please note that new requirements or changes will be accommodated within a Change Request (CR) scope. Additional charges, at a rate based on actual man-days, will be applied.

-

Expectations & Dependencies from the Department

- Clear Requirements: The EPCO Department is expected to designate a Single Point of Contact for the engagement. The EPCO Department is expected to provide clear and detailed requirements for the solution development.
- Regular Communication: The EPCO Department is expected to maintain open and regular communication with the project manager. This involves providing feedback, answering questions, and addressing any concerns promptly throughout the project lifecycle.
- Timely Decision-Making: The EPCO Department is expected to make timely decisions regarding project priorities, scope changes, and approval of deliverables. Delays in decision-making can impact project timelines and budgets.
- Resource Allocation: The EPCO Department is expected to allocate necessary resources to support the requirement-gathering phase for each module. This may involve providing access to subject matter experts.
- Quality Assurance: The EPCO Department is expected to participate in quality assurance activities, such as user acceptance testing (UAT), to ensure that the software meets their requirements and quality standards before deployment.
- Adherence to Deadlines: The EPCO Department is expected to provide timely feedback, review deliverables, and grant necessary approvals to ensure the project stays on track.
- Mobile App: The mobile app will be developed for the wetland survey. After the completion of the survey, MPSEDC will not be able to provide further support.
- Ramsar site – Bhoj Wetland boundary will be provided by EPCO in .shp format.

MPSEDC & EPCO Department Details:

The following professionals have been nominated as the key point of contact for this project.

From Department Name: -

Name of the Person:

Designation:

Contact No:

Email Address:

From MPSEDC: -

Name of the Person: Dr. Sandeep Goyal

Designation: Executive Director

Contact No: 09425370195

Email Address: sgoyal@mp.gov.in

Bank Details of MPSEDC		
1	Vendor Name (in Capital Letter)	MADHYA PRADESH STATE ELECTRONICS DEVELOPMENT CORPORATION LIMITED
2	Vendor Type	GOVERNMENT CORPORATION
3	Contact Person	Mohammed Muneer Qureshi
4	Address	47-A STATE IT CENTER ARERA HILLS BHOPAL NEAR BY KENDRIYA VIDHYALAY Bhopal 462011
5	State	MADHYA PRADESH
6	Mobile No.	0755-2518668,
7	Email Id	accounts@mpsdc.com
8	Bank Details	Bank of India
N	Account Holder Name	MADHYA PRADESH STATE ELECTRONICS DEVELOPMENT CORPORATION LIMITED
10	Bank & Branch Name	Bank of India Arera Hills Branch, Bhopal
11	Account No.	900710110006826
12	IFSC Code	BKID0009007
13	Pan No.	AABCM0089R
14	GST No. Certificate to be attached	23AABCM0089R1Z5

Payment Details		
1	Name of Department	
2	Purpose	
3	MPSEDC Letter Ref. No.	
4	UTR No.	
5	Date of Payment	